

GRAINGUARD · REQUIREMENTS

GrainGuard requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Design for construction (GGD-DDR-003); R13 reported against the value-engineering target (Amish, 2026-10-01); R7 margin from the modelled antenna
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R4 restated with the plenum probe in the base kit; R13 status records the accepted \$6.50 overrun; status from GGD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 decisions (GGD-DDR-001); R13 redefined per bin with the receiver per farm; R4 plenum EMC basis; reference case corrected; status from GGD-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

GrainGuard requirements

These requirements were checked by calculation at TRL 3 in GGD-CAL-001. Targets are not yet validated with farmers and will be revised after co-design visits (see GGD-PRB-001). One requirement is not met (R8, battery recovery), three are at risk (R2, R6, R9) and two cannot be verified at TRL 3 (R3, R10); Table 2 gives the status of each. Version 0.3 recorded the decisions in GGD-DDR-001: R13 is redefined per bin with the farmhouse receiver costed per farm (D1), and R4 states how the plenum-air EMC is found (D3, D9). Version 0.4 records GGD-DDR-002: the plenum probe is in the base kit, so R4 uses the measured plenum temperature and is met on paper, and the \$6.50 cost overrun it causes is accepted. Version 0.5 records the budget Amish approved on 2026-09-26. Version 0.6 follows the design for construction (GGD-DDR-003) and Amish's 2026-10-01 instruction that `budget_usd` is a hypothetical value-engineering target, not a limit: R13 is reported against that target. Value-engineering target: USD 260. Estimated cost of the constructable design: USD 389.50 per bin (USD 129.50 over the target).

The **reference case** is a 5.49 m (18 ft) diameter corrugated steel bin with a 5.6 m eave above the pad, a full perforated floor 0.4 m above the pad and grain level with the eave (5.2 m deep): about 88.8 t (3,493 bu) of shelled corn stored at a target of 15.0 % moisture, wet basis, in the US Midwest (GGD-DDR-001, D11), with one existing aeration fan giving about 0.330 m³/s (699 cfm, 0.2 cfm/bu).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Grain temperature profile	6 or more levels on one center cable, 1.0 m or less apart; ± 0.3 °C; one reading every 10 min	Sensor datasheet; geometry check in the model
R2	Grain moisture estimate from interstitial air	Within ± 0.8 percentage points, wet basis, for interstitial RH of 20 % to 75 % and grain at 0 °C to 30 °C, from readings taken 6 h or more after the fan stops	EMC error analysis with sensor tolerance; later comparison with oven or meter samples
R3	Ambient air measurement	± 0.3 °C and ± 2 % RH inside a radiation shield; sampled every 5 min; decisions on 30 min averages	Datasheet and shield error estimate
R4	Fan decision rule	Automatic mode runs the fan only when the plenum-air EMC (from the ambient humidity ratio and the plenum temperature, measured by the plenum probe in the fan transition, BOM item 14) is at or below the target moisture (hold mode), or when plenum air is 5 °C or more below the mean grain temperature and its EMC is no more than 1.0 point above target (cool mode); minimum run 30 min, minimum off 15 min, 4 starts per hour or fewer	Logic review against extension guidance; later simulation over a season of weather data
R5	Fail-safe control	In automatic mode the fan stops within 60 s of controller power loss or when sensor data are more than 30 min old; the hand position of the hand-off-auto switch runs the fan with GrainGuard removed; fan running is confirmed by current within 30 s and a mismatch raises an alarm	Circuit review; later bench test
R6	Heating alert	Alarm at the farmhouse within 15 min when any pod rises 3 °C or more above its 7-day minimum with the fan off, or sits 5 °C or more above the bin median	Logic review; later heated-core test
R7	Radio link	Bin to farmhouse over 1 km or more with one farm building in the path; 95 % or more of packets delivered	Link budget; later site survey
R8	Power autonomy	5 days or more with no solar input at -20 °C with the fan running continuously; recovers from 50 % charge to full in 3 winter days of 1.5 peak sun hours	Power budget
R9	Environment	Operate at -30 °C to +50 °C; controller IP66; pods dust-tight (IP6X) with a vapor-permeable membrane; survive one phosphine fumigation per season	Datasheets and design review; later exposure test
R10	Mechanical strength	Rope minimum breaking load 4 times or more the design pull-down force of 2.5 kN (estimate); roof hanger rated by the bin maker for the design force	Force estimate; bin maker's data
R11	Install and service without grain entry	Installed with the bin empty, from the roof manhole and floor under a confined-space and fall-protection procedure; no GrainGuard task needs anyone in a bin that holds grain	Installation sequence review
R12	Electrical isolation	Controller, cable and pods at 15 V DC or less; the relay kit is the only mains-connected item, installed by a licensed electrician, with 2.5 kV or more isolation between control input and contacts	Circuit review; relay datasheet
R13	Affordable		

ID	Requirement	Target	Verification (TRL 3 or later)
		Parts cost per bin reported against the value-engineering target of USD 260 (a hypothetical control target, not a limit; Amish, 2026-10-01), with the farmhouse receiver (one per farm) costed per farm and excluded; priced options excluded	Priced BOM (bom/ bom.csv)
R14	Local data	All readings and fan hours stored on the farm; no cloud account needed	Architecture review

Table 2. Status at TRL 3 (GGD-CAL-001 v0.3, Table 3). Not met items first.

ID	Status at TRL 3	Basis
R8	Not met	Autonomy 7.7 days with the fan continuous and a 3 mA relay input (met); refill from 50 % takes 7.3 days with the 10 W panel (not met); a panel of about 19 W would meet it (open, GGD-DDR-001 O2)
R2	At risk	Sensor-only error 0.23 points (SHT45) and 0.42 points (SHT40) worst case over 20 % to 75 % RH, 0.81 at the SHT40 maximum tolerance; equation fit error, drift and membrane lag not included
R6	At risk	Alarm latency about 13 min (met); only the center core, 3.3 % of the cross-section, is watched
R9	At risk	AGM electrolyte may freeze near -25 °C at 50 % charge; phosphine protection unverified
R3	Not verifiable at TRL 3	SHT45 meets the target; the radiation error of the shield needs a test
R10	Not verifiable at TRL 3	Rope 8 times the 2.5 kN design force (estimate 2.08 kN); the roof hanger rating comes from the bin maker
R4	Met	Rule met by design; fan heat measured by the plenum probe within ± 0.51 °C, so the plenum EMC is known within about ± 0.5 points (GGD-CAL-001, B10)
R1	Met	Six pods at 0.94 m pitch; ± 0.1 °C (SHT45) and ± 0.2 °C (SHT40) typical
R5	Met	Normally open relay, stale-data timeout and hand-off-auto switch
R7	Met	16.9 dB margin at 1 km past one farm building with the antenna on its mast bracket (plane-earth model); 24-byte packets fit the 400 ms dwell limit
R11	Met	Empty-bin installation sequence in GGD-PRC-001
R12	Met	12 V system with charging clamped at 15.0 V; optically isolated relay input
R13	Over the value-engineering target	USD 389.50 per bin for the constructable design (GGD-DDR-003), USD 129.50 over the USD 260 target; receiver USD 22.00 per farm. The concept kit was USD 256.50
R14	Met	Farmhouse receiver stores data locally

■ Assumptions

- Grain properties: shelled corn at 721 kg/m³ (25.4 kg per 0.0352 m³ bushel) and a specific heat of about 2.0 kJ/(kg·K) at 15 % moisture.

- EMC from the ASABE D245.6 modified Henderson equation for corn; accuracy of RH-based moisture prediction as reported by [USDA ARS](#).
- The fan warms the air by its input power over the air's heat capacity rate: about 1.0 °C for a 0.40 kW fan, 0.5 to 2.7 °C for 0.2 to 1.1 kW (GGD-CAL-001, section A).
- Cooling front time of 75 to 119 h at 0.2 cfm/bu, from extension guidance and a sensible-heat balance.
- Design pull-down force of 2.5 kN for a thin 6 mm cable is kept; GGD-CAL-001 estimates about 2.08 kN with an unloading overpressure factor of 2.0. Forces up to 4.7 kN were measured on the largest commercial cable during unloading ([Illinois Experts](#), [ASABE](#)); thinner cables see less force, but this is unverified.